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Department of Electronics & Communication Engineering | GATE Preparation Library

Electromagnetics (EMFT) — Core Formula Sheet

Maxwell's Equations, Plane Wave Impedances, Transmission Lines, Waveguides & Antennas

GATE Section: Section 8 • **Marks:** 8 - 10 Marks • **Core Constants:** eta0 = 377 Ohms, c = 3x10^8 m/s

MAXWELL'S EQUATIONS & BOUNDARY CONDITIONS
- **Differential Form:** div D = rho_v | div B = 0 | curl E = -dB/dt | curl H = J + dD/dt.
- **Boundary Conditions:** Tangential E1t = E2t | Normal B1n = B2n | Tangential H1t - H2t = K x an | Normal D1n - D2n = rho_s.
- **Quarter-Wave Line:** Zin = Z0^2 / ZL.

SECTION: PLANE WAVE PROPAGATION & POYNTING VECTOR

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Wave Propagation Constant	**gamma = alpha + j*beta = j*omega*sqrt(mu * epsilon_c)** **vp = omega / beta = 1 / sqrt(mu * epsilon)**	alpha = attenuation constant (Np/m) beta = phase constant (rad/m) = 2*pi / lambda	In lossless medium: alpha = 0, beta = omega*sqrt(mu*epsilon), vp = c / sqrt(mu_r*epsilon_r).
Intrinsic Wave Impedance	**eta = sqrt(mu / epsilon_c)** **Free Space: eta0 = sqrt(mu0 / epsilon0) ~ 120*pi ~ 377 Ohms**	E / H = eta in magnitude E, H, and wave vector k are mutually orthogonal	In good dielectric: eta is real. In good conductor: eta = sqrt(omega*mu / sigma) * exp(j*pi/4).
Skin Depth (Good Conductors)	**delta = 1 / alpha = sqrt(1 / (pi * f * mu * sigma))** **alpha = beta = sqrt(pi * f * mu * sigma)**	delta = depth at which wave decays to 1/e (~37%)	At microwave frequencies, current flows entirely within a few micrometers of conductor surface.
Poynting Vector & Power	**S = E x H (Instantaneous, W/m^2)** **S_avg = 0.5 * Re{E x H*} = (|E0|^2 / (2 * eta)) * a_k**	Time-average power density flowing in direction of propagation a_k	Total radiated power = closed surface integral of S_avg • dS.
Reflection & Transmission (Normal)	**Gamma = (eta2 - eta1) / (eta2 + eta1)** **tau = 2 * eta2 / (eta2 + eta1) = 1 + Gamma**	Gamma = reflection coefficient, tau = transmission	Standing Wave Ratio: SWR = (1 + |Gamma|) / (1 - |Gamma|).

SECTION: TRANSMISSION LINES & IMPEDANCE MATCHING

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Lossless Parameters	**Z0 = sqrt(L / C) Ohms** **beta = omega * sqrt(L * C)** **vp = 1 / sqrt(L * C)**	L in H/m, C in F/m	Distortionless condition: R / L = G / C -> alpha = sqrt(R*G), beta = omega*sqrt(L*C).
Input Impedance Equation	**Z_in(l) = Z0 * [(Z_L + j*Z0*tan(beta*l)) / (Z0 + j*Z_L*tan(beta*l))]**	l = line length, beta = 2*pi / lambda	Quarter-wave line (l = lambda/4): Zin = Z0^2 / ZL. Half-wave line (l = lambda/2): Zin = ZL.
Voltage Reflection Coeff	**Gamma_L = (Z_L - Z0) / (Z_L + Z0)** **Z_in = Z0 * [1 + Gamma(l)] / [1 - Gamma(l)]**	Gamma(l) = Gamma_L * exp(-2*j*beta*l)	|Gamma| is constant along a lossless line; only its phase rotates by 2*beta*l radians.

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Short & Open Circuit Stubs	**Short (ZL = 0): Zin = j * Z0 * tan(beta*l)** **Open (ZL = inf): Zin = -j * Z0 * cot(beta*l)**	Acts as pure reactive element (inductor or capacitor)	Length < lambda/4: Shorted line is INDUCTIVE, open line is CAPACITIVE.

SECTION: WAVEGUIDES, RESONATORS & ANTENNAS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Waveguide Cutoff Frequency	**fc_mn = (c / 2) * sqrt((m/a)^2 + (n/b)^2)**	a = width, b = height (a > b)	Dominant mode: TE10 mode with fc_10 = c / (2*a). Guide wavelength: lambda_g = lambda0 / sqrt(1 - (fc/f)^2).
Wave Velocities in Guide	**vp * vg = c^2** **vp = c / sqrt(1 - (fc/f)^2) > c** **vg = c * sqrt(1 - (fc/f)^2) < c**	vp = phase velocity (exceeds c!) vg = group velocity (information transport speed)	Wave impedance: Z_TE = eta0 / sqrt(1 - (fc/f)^2), Z_TM = eta0 * sqrt(1 - (fc/f)^2).
Infinitesimal Dipole Radiation	**R_rad = 80 * pi^2 * (dl / lambda)^2** **Directivity D = 1.5 (1.76 dBi)**	dl << lambda (Hertzian dipole)	Radiation resistance is small; poor radiation efficiency for small lengths.
Half-Wave Dipole Antenna	**R_rad = 73 Ohms** **Input impedance ~ 73 + j*42.5 Ohms** **Directivity D = 1.64 (2.15 dBi)**	Length l = lambda / 2 Sinusoidal current distribution	Effective aperture: Ae = (lambda^2 / 4*pi) * D = 0.13 * lambda^2.
Friis Transmission Equation	**P_r / P_t = G_t * G_r * (lambda / (4 * pi * R))^2**	Pt = transmit power, Pr = received power R = separation distance, Gt, Gr = antenna gains	Received power decays as 1/R^2 (free space path loss).